

Action code: WHEN CONVENIENT

New ALCU 2R
replacing the existing ALCUSL2025-766/ANRE
April 2025**Concerns**

Owners and operators of MAN B&W two-stroke marine combustion engines.
Type: MC/MC-C mounted with Alpha lubricators

Summary

The ALCU 2R is a new, future-proof version of the Alpha Lubricator Control Unit, designed to replace the original ALCU with improved functionality and support for real-time monitoring and optimal engine performance.

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Dear Sir or Madam

The Alpha Lubricator Control Unit (ALCU) has been in service for more than two decades with great success. Due to the end of life of the components for the ALCU, we have developed a new, future-proof version of the ALCU called ALCU 2R.

The ALCU 2R is being introduced as an electronic solution to control the engine cylinder lubrication system based on real-time monitoring. As a successor to the well-known ALCU, the ALCU 2R offers the possibility to adjust the cylinder lubrication process to be optimal at all load points to ensure balanced engine performance regardless of load, load changes, and fuel variations.

The new ALCU 2R system is available as a retrofit solution and will be supported in the future. The ALCU 2R will be mounted in the existing cabinet of the current version of ALCU.

The ALCU 2R is a PLC-based solution that utilizes a set of standardized building blocks that can be assembled and programmed in various, but finite, configurations.

For more details on the new ALCU 2R, see page 2.

If you are interested in a last-time buy for the ALCU parts, please contact your local sales office: Primeserv-CPH@man-es.com

Yours faithfully



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Why did we design a new ALCU?

The following parts from the original ALCU will become obsolete: Master Control Unit (MCU), Switch Board Unit (SBU), Backup Control Unit (BCU), and HMI panel. These components will be offered for a limited time as long as MAN Energy Solutions has stock availability.

The ALCU 2R is based on a standardized industrial controller. The controller uses a single multiconnector for almost all signals and is therefore easy to replace. For engines with up to 7 cylinders one controller is sufficient, but for engines with 8 cylinders or more two controllers are needed.

The new ALCU 2R system lubricates in the same way as the existing system. Alarm handling and lubrication modes will remain the same. Some alarms have been omitted due to the new concept, and new alarms have been added to cover the new system design. Additional functionality is included by means of earth failure detection.

Difference between the new and the existing system:

The main difference is the backup philosophy. The existing system has a separate BCU controller that can maintain lubrication (untimed) if the MCU controller is damaged. The new system has backup by easy exchange of the single main controller.

Another significant difference is the alarm system. Hardwired alarms are reduced to one common alarm and a slowdown signal. Other hardwired alarms have been omitted as the BCU/SBU system no longer exists. Power fail alarms are implemented in the software, and two new local alarms are introduced.

The HMI panel layout is the same as in the existing system; however, the LAMP-TEST button is changed to an operating mode indicator (timed or untimed mode).

Additionally, the new system has some important add-ons to the functionality. The ALCU 2R system comes as standard with a DC/DC converter and Insulation Resistance Monitor, which means that the new system is less prone to failure if an external isolation problem is encountered. Furthermore, an isolation failure in the ALCU system will not affect the ship system. The new system has an internal isolation monitor system which will raise an alarm if the isolation level becomes too low. A new menu is implemented where the overall isolation level can be monitored on the HMI panel during operation.

The function of the new ALCU 2R:

- Within **Normal** operation mode, the controller performs the same function as the MCU in the existing system by

providing **Timed Injection**: In the ring pack based on the angle measurements from the shaft encoder. The injection of oil is timed according to the crankshaft position.

Supervision of the injection is performed the same way as the existing system and will raise the alarm and eventually slow down if lubrication can't be maintained.

- Within the **Backup** operation mode, the controller performs the same function as the BCU in the existing system by providing **Untimed** Injection based on the engine speed measured by the BCU pickups. As the injection of cylinder oil is performed randomly, an additional amount is added to compensate for the missing timing. The untimed lubrication mode will be activated in the event the encoder system is failing, either the TDC marker or trigger signals are lost. Untimed lubrication will continue to lubricate the cylinders according to the engine speed measured by either the TDC signal from the encoder if available or signals from the two BCU pickups.

In case of failure of the main controller:

As the existing BCU has been omitted in the new design, the entire ALCU 2R main controller will have to be exchanged if it is damaged or failing. In case of main controller failure, the main controller can be replaced by reducing the main engine to low-load operation and maintaining manoeuvring speed for up to 30 minutes. In this period, it is safe to change the controller, which takes less than 10 minutes. Software and engine-specific configuration will automatically be uploaded to the spare controller when installed.

A spare main controller should always be kept in stock on board the vessel.

Installation/application requirements: ALCU 2R is available for all MC/MC-C engine types equipped with the Alpha Lubricator lubrication system. Installation must be done by service personnel from MAN PrimeServ and will take one day.

Product launch timeline: ALCU 2R is expected to be available in second half of 2025.

Certification and approvals: ALCU 2R is approved by classification societies as a replacement for the existing ALCU system.

Delivery projections: The market will be informed in due time regarding pricing structures, delivery times, and other relevant product information.

Questions regarding the ALCU 2R should be directed to your local sales office or PrimeServ DK Retrofit & Upgrade department at: RetrofitDK@man-es.com